

ADDENDUM A

World Foundation Models

Closing the Synthetic-Reality Gap in v0.1

Companion to:
Propagation Identity — The Missing Primitive for Physical AI (v0.1.0, April 2026)

April 30, 2026 — Addendum v0.1.0a

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Preface — What This Addendum Adds, and Why

Version 0.1.0 of the Propagation Identity whitepaper, dated April 2026, defined a four-layer signing chain for embodied artificial intelligence: operator → model → runtime → actuator. The four layers were chosen as the minimal set sufficient to span the decision-to-action boundary that motivates the paper. In the eighteen months prior to v0.1.0, however, a fifth class of model has crystallized into a distinct, commercially-deployed category: the world foundation model (WFM).

World foundation models are generative models that produce synthetic physical reality — interactive 3D environments, photo-real video sequences, physics-aware simulations — from text, image, video and motion inputs. As of April 2026, three platforms have reached broad deployment: NVIDIA Cosmos (2 million downloads, Cosmos 3 announced 16 March 2026), Google DeepMind Genie 3 (public release 29 January 2026 to AI Ultra subscribers), and World Labs Marble (commercial API, \$20–95 per month). Yann LeCun’s AMI Labs (€500M, €3B valuation, Paris, 2025) is the European entrant. NVIDIA’s adopter list overlaps almost entirely with the platform table in §4.3 of v0.1.0: 1X, Agility Robotics, Figure AI, XPENG, Agile Robots, Hillbot, Skild AI, AGIBOT, NEURA Robotics, LG Electronics, Foretellix, Uber, Waabi.

The four-layer signing chain of v0.1.0 does not contain a slot for the world foundation model. This is, on reflection, a structurally significant gap. A VLA policy that was trained on Cosmos-generated synthetic data is an AI safety component whose behaviour was shaped by a model that the operator never directly invokes; under Article 23 of the Machinery Regulation, a behaviour change in that upstream model is a behaviour change in the safety component. A planner that queries Cosmos Reason 2 mid-execution for spatial reasoning is invoking a world model at inference time, on a sub-second cadence, with consequential effects on the action chain. Neither case is represented in v0.1.0.

This addendum closes the gap. It does so additively rather than by rewriting v0.1.0. The four-layer chain remains correct as a minimum; the addendum extends it to a parametric N-layer chain, with the world foundation model occupying a new sub-layer L1.5 in the provenance stack and a new optional position in the signing chain. The addendum specifies the schema extensions required of MobyDB, the signing-chain extensions required of the propagation primitive, and the policy questions that arise when a regulated deployment depends on a model whose weights are governed by a non-EU provider. The addendum is intended to be read alongside v0.1.0; references to specific sections of v0.1.0 are given in the form §X.Y.

§4.4 World Foundation Models — A Fifth Model Category

Sections §4.1 through §4.3 of v0.1.0 enumerated nine commercially-deployed humanoid and industrial-arm platforms, identifying the VLA stack of each. The VLA stack — vision-language-action models such as Gemini Robotics 1.5, Helix 02, Redwood, and the in-house stacks at Tesla, XPENG and Unitree — was treated as the model layer of the propagation chain. This section documents a model category that sits upstream of the VLA layer: the world foundation

model. The category was visible but not crystallized at the time v0.1.0 was drafted; in the four-month period December 2025 to March 2026 it has crystallized.

4.4.1 Definition and Scope

A **world foundation model** is a generative model that, given text, image, video or motion inputs, produces synthetic physical reality: video sequences, interactive 3D environments, or simulated sensor streams that exhibit physical consistency over time. The model is trained on large-scale real-world video data and is, in current implementations, autoregressive (generating frame-by-frame conditional on history) rather than producing an explicit 3D scene representation. The category is distinct from prior video-generation models (Sora, Veo, Runway), which produce passive sequences, and from prior 3D reconstruction techniques (NeRFs, Gaussian Splatting), which reconstruct existing scenes. A WFM produces a controllable, physics-aware environment that an agent can act in.

4.4.2 The Three Reference Platforms

Platform	Provider	Architecture	Deployment status (April 2026)
NVIDIA Cosmos 3	NVIDIA	Diffusion + autoregressive transformers; Predict / Transfer / Reason components	2M+ downloads; Open Model License; adopted by 1X, Agility, Figure, XPENG, Skild AI, NEURA, Uber, Waabi
DeepMind Genie 3	Google DeepMind	Autoregressive frame generation (no explicit 3D scene); 24 fps; ~1 minute memory	Public 29 Jan 2026 to AI Ultra subscribers (US); described as “stepping stone to AGI” by DeepMind
World Labs Marble	World Labs (Stanford-adjacent)	Hybrid 3D editor + generative fill	Commercial API since late 2025; \$20–95/month; Unreal-compatible export

The three platforms are not feature-equivalent. Cosmos is positioned for industrial physics consistency (robotics and AV training data); Genie 3 is positioned for creative exploration and agent training in diverse environments; Marble is positioned for production 3D-asset pipelines compatible with Unreal Engine. For the propagation primitive the differences matter operationally — schema, version-pinning, jurisdictional posture all vary by platform — but the structural argument is the same across all three.

4.4.3 Two Distinct Operational Positions in the Chain

A world foundation model can enter the propagation chain in two structurally distinct positions. Each has different signing requirements and different regulatory implications.

Training-time position (synthetic-data provenance)

In the training-time position, the WFM produces synthetic data on which a VLA policy is subsequently trained. The trained VLA is the model that operates in production; the WFM is no longer in the loop at runtime. NVIDIA Cosmos is most commonly used in this position: a robotics company synthesizes terabytes of physics-aware video using Cosmos, post-trains a VLA on the synthetic data, and deploys the VLA in production. The operational chain does not invoke Cosmos in production; the operational chain depends on Cosmos.

The regulatory consequence is direct. Under Article 23 of the Machinery Regulation 2023/1230, a change in the AI behaviour triggers re-conformity assessment. If NVIDIA releases Cosmos 3.5, and the operator's VLA is re-distilled from Cosmos 3.5 synthetic data, the safety component's behaviour has changed. Without a record of which WFM version produced the training data, the operator cannot demonstrate to a notified body which behaviour is being assessed. The training-time WFM is therefore part of the propagation provenance — but it is provenance of training, not of inference. It is recorded once per VLA release, not per action.

Inference-time position (runtime spatial reasoning)

In the inference-time position, the WFM is invoked during action execution. Cosmos Reason 2, released in February 2026, is explicitly designed for this position: it accepts visual inputs and text prompts and produces 2D/3D point localizations and bounding-box coordinates with reasoning explanations, with up to 256K context tokens. A planner that queries Cosmos Reason 2 mid-skill to resolve a spatial ambiguity is invoking the WFM as part of the model layer of the propagation chain. The WFM is no longer upstream; it is in the loop.

The regulatory consequence is parallel to the model-layer treatment in §8. Each invocation of the inference-time WFM is a discrete decision that influences the runtime plan. Each invocation must be signed by the WFM provider's key (or, in the absence of a provider-signed inference, by a runtime attestation of the queried output). The WFM-signature is a new payload field in the breadcrumb.

4.4.4 Structural Difference From the Existing Model Layer

The existing model layer (§8.2 of v0.1.0) signs the inference of the embodied-reasoning planner — the decision about what motor primitives to execute. The world-foundation-model layer signs a different question: what physical reality the planner believes it is operating in. The two are not equivalent. A planner can produce a correct signed inference about a hallucinated world; a world model can produce a correctly-signed world description that bears no relation to the physical environment. Layer 5 propagation must capture both.

Patches to Existing Sections of v0.1.0

This section specifies the additive changes required to integrate world-foundation-model attestation into the existing whitepaper sections. Each patch is small and additive; v0.1.0 reads consistently with or without the patches applied. Operators integrating the addendum should regard the patched version as v0.1.1; a fully-integrated v1.0 will absorb all patches into the body text.

Patch P1 — §2.2 Eight-Primitive Gap-Analysis Table

Add one row to the gap-analysis table of §2.2:

Primitive	Established Layer	Termination Boundary	What Remains Unsigned
WFM provider attestation (none in 2026)	World-model output	The generative output (video / scene)	Which WFM version produced this scene; which actions inside the scene mattered for downstream behaviour

Note: as of April 2026, no commercial WFM provider signs its outputs. NVIDIA, DeepMind and World Labs all publish their models under licenses that include version identifiers, but the inference outputs themselves are unsigned. This is the gap that motivates the proposed L1.5 layer below.

Patch P2 — §6.1 Provenance Stack: insert L1.5

Insert a new sub-layer L1.5 between L1 (media provenance) and L2 (build-time provenance):

Layer	Layer Name	Representative Primitives	Bound Object
L1	Media provenance	C2PA; Content Credentials	Media file (image, video, audio)
L1.5	World-model provenance (proposed)	This addendum; WFM version manifests; inference attestation hooks	Synthetic 3D environment; physics-aware video; runtime spatial inference
L2	Build-time provenance	SLSA; in-toto; Sigstore	Build artefact; release

Rationale. L1 (C2PA) is designed for 2D media intended for human consumption. WFM outputs are 3D environments and physics-aware sequences intended for AI consumption — they are not a special case of 2D media. The cryptographic primitive is similar (cryptographic provenance of a generative output), but the schema and the verification semantics differ. The

cleanest architecture is a sibling sub-layer at L1.5 that composes C2PA where appropriate and extends C2PA where not. The Layer 5 propagation primitive references L1.5 attestations via the same hook mechanism that references L1 (see §13 schema patch P5).

Patch P3 — §7.2 Property P1 (Composability) — extend

Property P1 in §7.2 currently states that the chain must span Layer 4 (software-agent primitives) and Layer 0 (physical-actuation primitives). Extend the property to span L1.5 also: a propagation chain in a deployment that uses a world foundation model must compose the L1.5 attestation in addition to L0 through L4. The verifier semantics are unchanged; only the set of primitives composed is extended.

Patch P4 — §8 Signing Chain — parametric N-layer

The four-signature chain of §8 is correct as a minimum. Extend the chain to be parametric in the number of layers: at deployment time, the operator declares the layers in the chain by certificate, and the verifier checks that the signatures correspond to the declared layers. Two new layers are recognized:

- `L_world_train` — training-time WFM provenance. Recorded in the operator delegation certificate (§8.2) and the model-layer certificate, not per-breadcrumb. References a WFM version manifest that includes provider, model name, version, weights hash, and license under which the model was used.
- `L_world_infer` — inference-time WFM invocation. Recorded per breadcrumb, in a new payload field. Each WFM call mid-action produces an attestation that is included by reference (hash) in the breadcrumb. The actual WFM output (which can be megabytes of video frames or 3D scene data) is stored in object storage, with the hash chained into the breadcrumb.

```
// Updated breadcrumb structure (additive)
{
  ... (all v0.1.0 fields preserved) ...

  // New L1.5 fields — optional, present when WFM is in chain
  "world_model_train": {                                // training-time provenance
    "provider": "nvidia",
    "model":    "cosmos-predict-2.5-2b",
    "version":  "2.5.0",
    "weights_hash": "blake3:e84f...",
    "license":  "NVIDIA Open Model License v1.0"
  },
  "world_model_infer": [                                // 0..N invocations during
    {                                                    // this action
      "provider":    "nvidia",
      "model":       "cosmos-reason-2",
      "version":     "2.0.1",
      "prompt_hash": "sha256:9a23...",
      "output_hash": "blake3:c7b1...",
      "output_uri":  "r2://obj/cosmos-out/...",
      "attestation": "signed-by-runtime-or-provider"
    }
  ]
}
```

Figure A.1 — Additive payload fields for world-foundation-model attestation.

Patch P5 — §13 MobyDB Schema — additive columns

Add the following columns to the `propagation_breadcrumbs` collection. All columns are nullable so existing v0.1.0 deployments are forward-compatible:

```
ALTER COLLECTION propagation_breadcrumbs
  ADD COLUMN world_model_train JSON,           -- L_world_train
  ADD COLUMN world_model_infer JSON[],         -- 0..N entries
  ADD COLUMN world_model_count INT DEFAULT 0; -- query helper
```

Patch P6 — §15 Comau Worked Example — clarify

The worked example of §15 should be re-read with the addendum in mind. Two clarifications are added:

- The VLA executing on the Comau workcell is, in 2027, plausibly Cosmos-distilled. The training-time WFM provenance therefore appears in the operator's standing delegation certificate, not per-breadcrumb. Re-distillation from a new Cosmos version is a delegation update, recorded as a delegation rotation in the operator's certificate chain.
- If the planner queries Cosmos Reason 2 at runtime to resolve a spatial ambiguity in the workcell, the inference-time WFM hook fires. The example breadcrumb of §15.3 would carry an entry under `world_model_infer` naming the model version and hashing the query/response.

Patch P7 — §18 Open Questions — add §18.7**§18.7 Provider attestation for world foundation models**

As of April 2026, no commercial WFM provider signs its inference outputs. NVIDIA, DeepMind and World Labs publish version identifiers and license terms, but inference outputs are unsigned. The cleanest path to provider attestation is for the WFM provider to sign each inference with a provider key rooted in their own attested infrastructure; the propagation breadcrumb then references the provider attestation by hash. This requires WFM providers to add signing infrastructure to their inference APIs. The case for them to do so is the same case the present paper makes for embodied-AI providers: regulatory exposure under the AI Act and the Machinery Regulation. The companion paper announced in §19.5 of this addendum is the proposed vehicle for that conversation.

Patch P8 — §19 Standardization — add §19.5 (NVIDIA / DeepMind / World Labs)**§19.5 Coordination with World Foundation Model Providers**

v0.1.0 enumerated coordination targets at W3C, OASIS C2PA, ISO/IEC SC 42, CEN-CENELEC JTC 21, the EU AI Office, and NIST. World foundation model providers are an additional coordination class:

- NVIDIA (Cosmos team) — open model license already published; commercial interest in defining the de-facto attestation standard for synthetic data used in regulated training.
- Google DeepMind (Genie / Project Genie team) — interactive WFM for agent training; AI Ultra subscriber distribution model; positioned within Google AI structure that already houses Gemini Robotics.
- World Labs (Marble team) — commercial API; Stanford-adjacent; smaller footprint but the cleanest licensing posture.
- AMI Labs (Yann LeCun) — Paris-based; €500M raise December 2025; physics-first agenda; the European interlocutor of choice for an EU-rooted standardization effort.

A companion paper, World Foundation Model Attestation — Extending Propagation Identity to Synthetic Reality, is in preparation for Q3 2026 publication, with co-authoring sought from the four providers above. Outreach plan and target schedule are documented in the operational intelligence brief that accompanies this addendum.

Appendix D — World Foundation Model Glossary and Reference Versions

D.1 Glossary

Autoregressive frame generation. A model architecture in which each output frame is generated conditional on previously generated frames and a running context, rather than being decoded from an explicit 3D scene representation. Genie 3 uses this architecture; Cosmos Predict and Transfer use a hybrid approach (diffusion + autoregressive).

Cosmos Predict. NVIDIA Cosmos component for generating predictive video worlds from text, image or video prompts. Cosmos-Predict2.5-2B as of April 2026, available via Hugging Face.

Cosmos Reason. NVIDIA Cosmos component for vision-language reasoning over generated and real video. Cosmos Reason 2 (February 2026) adds 2D/3D point localization, bounding-box coordinates, and 256K input-token context.

Cosmos Transfer. NVIDIA Cosmos component for sim-to-real visual translation; pairs with NVIDIA Omniverse and Isaac Sim. Cosmos Transfer 2.5 as of December 2025.

Genie 3. Google DeepMind world model. Public release 29 January 2026 to AI Ultra subscribers in the United States. Real-time interactive 3D environments at 24 fps, ~60 second sessions, ~1 minute memory horizon.

Inference-time WFM. Use of a world foundation model during action execution to resolve spatial ambiguity, reason about scene composition, or augment perception. Cosmos Reason 2 is the canonical example.

Marble. World Labs commercial WFM API. Hybrid 3D editor + generative fill; export compatible with Unreal Engine. Pricing \$20–95/month.

Open Model License. NVIDIA-issued license under which Cosmos models are distributed. Permits commercial use; requires attribution; does not yet specify attestation requirements.

Synthetic-data provenance. The chain of evidence connecting a downstream model (typically a VLA) to the world-foundation-model run that produced its training data. Required by Article 23 of Machinery Regulation 2023/1230 when WFM-distilled VLAs are deployed as safety components.

Training-time WFM. Use of a world foundation model to generate synthetic data on which a downstream model is subsequently trained. The WFM is not in the production loop; the production model depends on the WFM at training time only.

World Foundation Model (WFM). A generative model trained on large-scale real-world video data, capable of producing synthetic physical reality (interactive 3D environments, physics-aware video, simulated sensor streams). NVIDIA, DeepMind and World Labs are the three commercially-deployed providers as of April 2026.

D.2 Reference WFM Versions, April 2026

Model	Version	Released	Distribution
Cosmos 3 (platform)	3.0	16 March 2026	NVIDIA Open Model License
Cosmos Predict 2.5	2.5.0	December 2025	Hugging Face; NGC
Cosmos Transfer 2.5	2.5.0	December 2025	Hugging Face; NGC
Cosmos Reason 2	2.0.x	February 2026	NGC
DeepMind Genie 3	3 (public)	29 January 2026	AI Ultra subscription (US)
World Labs Marble	GA	late 2025	Commercial API; \$20–95/month
AMI Labs (forthcoming)	—	expected H2 2026	Not yet released; €500M raise Dec 2025

D.3 Adopter Overlap With v0.1.0 Platform Table

The following platforms appear both in §4.3 of v0.1.0 and on the public adopter list of NVIDIA Cosmos:

- 1X — confirmed Cosmos adopter; in-house VLA stack post-trained on Cosmos data.
- Agility Robotics — confirmed Cosmos adopter; Digit deployments at GXO.
- Apptronik — adjacency to Cosmos via the broader NVIDIA Isaac stack.
- Figure AI — confirmed Cosmos adopter; Helix VLA in production.
- XPENG — confirmed Cosmos adopter; XPENG Physical AI stack.
- Skild AI — confirmed Cosmos Transfer adopter for synthetic data augmentation in Isaac Lab.
- Hillbot — Cosmos adopter for terabyte-scale 3D environment generation.
- AGIBOT, Mentee, NEURA Robotics — building humanoids with Cosmos + Isaac Sim + Isaac Lab + GR00T N.

The conclusion is that the §4.3 table understates the model layer: most platforms in the table operate on a stack that includes a world foundation model in the training-time position, even where the production VLA is in-house. The propagation chain in v0.1.0 captures the production VLA but not the training-time provenance from which it derives. Patch P4 closes this gap.

D.4 References

- NVIDIA Corporation — “NVIDIA Cosmos World Foundation Models — Open Model License,” NGC catalog, accessed April 2026.

- NVIDIA Corporation — “Cosmos 3 platform announcement,” 16 March 2026, nvidianews.nvidia.com.
 - NVIDIA Corporation — “Cosmos Reason 2 technical blog,” 9 February 2026 update, developer.nvidia.com.
 - Google DeepMind — “Genie 3: A new frontier for world models,” 5 August 2025; “Project Genie public release,” 29 January 2026.
 - World Labs Inc. — Marble product documentation, accessed April 2026, worldlabs.ai.
 - Sifted — “Yann LeCun raising €500m at €3bn valuation,” reporting on AMI Labs incorporation, December 2025.
 - Cosmos World Foundation Model Platform for Physical AI — NVIDIA Research, [arXiv:2501.03575](https://arxiv.org/abs/2501.03575).
 - World Simulation with Video Foundation Models for Physical AI — NVIDIA Research (Cosmos-Predict2.5), [arXiv:2511.00062](https://arxiv.org/abs/2511.00062).
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Concluding Note to the Addendum

This addendum closes the world-foundation-model gap in v0.1.0 of the Propagation Identity whitepaper. It does so by extending the four-layer signing chain to a parametric N-layer chain, by inserting a new sub-layer L1.5 in the provenance stack of §6, by extending the MobyDB schema with two nullable columns, and by adding a coordination class to the standardization plan of §19. The changes are additive; v0.1.0 reads consistently with or without the addendum applied; v0.1.1 of the whitepaper is v0.1.0 plus this addendum.

A companion paper, World Foundation Model Attestation — Extending Propagation Identity to Synthetic Reality, is in preparation for Q3 2026 publication. The companion paper is the vehicle for engaging NVIDIA, DeepMind, World Labs and AMI Labs as co-authors on the standardization of WFM attestation. The outreach plan and target list are in the operational intelligence brief published alongside this addendum.

“The journey is the proof — even when the journey is synthetic.”

— Addendum A, GNS Foundation, April 2026

— END OF ADDENDUM —